

ERIK TAYLOR, PhD

AI Systems Architect & ML Engineer | Agentic Systems, Dev Tooling & Research | Evaluation & Risk
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PhD engineer with 10+ years taking applied AI from research to shipped products. Architect of agentic LLM systems, the developer tooling that makes them buildable, and the evaluation harnesses that prove they work — most recently through agentic coding workflows, as human-in-the-loop director of AI-assisted engineering. Prior: production computer vision at Sony, frontier-model evaluation at Snorkel AI, and >\$2M in NIH-funded computational imaging research, with 50+ peer-reviewed publications. A practice built on measurement, reliability, and safe deployment in regulated environments.

PROFESSIONAL EXPERIENCE

AI Research Lead & ML Engineer

Oct 2024 – Present

Taylor ML R&D | San Francisco, CA

- **Agentic AI Systems:** Architected apply-daemon (Apache-2.0, open-source) — an LLM-orchestrated agentic pipeline that ingests roles and triages through a multi-stage cascading-LLM workflow (extract → judge → evaluate), routing scored matches to a Slack human-in-the-loop layer where one reaction triggers deep-research resume synthesis. Cost-optimized via nano-tier model inference with on-demand escalation to frontier models.
- **Developer Tooling for Agentic Engineering:** Designed and released dev-plan (MIT, open-source) — a Claude Code skill that converts planning discussions into scored, dependency-ordered development plans via a weighted implementation-priority model. Part of a broader skill suite covering cross-plan dependency coordination and structured ideation; published the design rationale and ran a controlled comparison against the built-in planning baseline.
- **Agentic Development Practice:** Build and ship production repositories primarily through agentic coding workflows — Claude Code (CLI and chat) for architecture, implementation, debugging, and iteration. Operate as the human-in-the-loop director of AI-assisted engineering for apply-daemon, dev-plan, and the autonomous portfolio manager.

AI Evaluation Specialist · Contract

Feb 2025 – Aug 2025

Snorkel AI | Remote

- **Frontier Model Risk & Reliability:** Authored advanced mathematical and graduate-level test cases designed to expose reasoning failure modes in frontier LLMs. Findings directly informed RL reward signals and alignment strategy for next-generation models and produced technical insights for production-readiness risk management.

Senior Algorithm Engineer

Jul 2022 – Oct 2024

Sony Electronics | San Jose, CA

- **Production Computer Vision at Scale:** Led end-to-end development of a two-stage vision pipeline shipped to phones and VR headsets — a sub-1 MB CNN with cascaded inference across sensor and host. Owned data curation, multi-GPU distributed training (DDP), INT8 quantization, and latency optimization (~30 FPS), securing a major B2B design win with a Fortune 500 customer.
- **Foundation-Model Roadmapping & AI Architecture:** Drove the technical roadmap for next-generation AI features by benchmarking SOTA vision and language foundation models — including a working prototype of open-vocabulary detection (OWL-ViT, CLIP) for promptable on-device perception — against performance, safety, and deployment constraints. Translated architectural trade-offs into product strategy.
- **Cross-Functional Program Recovery:** Orchestrated a high-stakes response to a sudden data policy change by bridging legal, business, and academic partners to co-develop a compliant dataset. Unblocked a next-generation AI hardware initiative and rescued a multi-month delivery timeline for a flagship pre-market product.

- **Model Validation & Safety Criteria:** Defined evaluation protocols for edge AI products targeting regulated markets, establishing internal standards for performance, bias, and deployment constraints aligned with responsible AI principles.

Research Assistant Professor, Medical Imaging AI

Apr 2019 – May 2022

University of New Mexico – Medical Physics & Radiology | Albuquerque, NM

- **Technical Program Leadership:** Led the AI strategy and computational architecture for a multi-disciplinary center with active NIH grant funding. Governed imaging data pipelines and model development across cardiac, neuro-oncology, and traumatic brain injury studies.
- **Biomedical Data Architecture (Journal Cover Feature):** Rescued a stalled traumatic brain injury study by architecting a custom deep learning framework (U-Net variant) capable of processing noisy, unstandardized biomedical data. Validated the architecture against baselines, earning a cover feature in the Journal of Neurotrauma (2023).
- **Cross-Functional Clinical Integration:** Bridged engineering researchers and clinical stakeholders (Radiology, Neurology, Oncology) to translate AI research into actionable clinical validation studies. Deployed HIPAA-compliant data handling, IRB-governed protocols, and clinical data governance across multi-site imaging studies.

NIH Research Fellow, Cardiovascular MRI

Jul 2016 – Mar 2019

Boston University | Boston, MA

- **Translational Imaging Research:** Developed and validated MRI acquisition and analysis methods for cardiovascular disease, combining physics-informed feature engineering with deep learning, and establishing a framework for choosing deep learning vs. classical methods by efficiency and clinical fit.

SELECTED PROJECTS

- **Autonomous Portfolio Manager (private).** Market-calibrated LLM agents reason over tools (financials, web search, filings retrieval) to generate investment theses; autonomous execution via Alpaca with risk guardrails and paper/live modes; 8-tab Streamlit dashboard for operator oversight. Evaluation platform with verified-reward decision tracking – +23.4% excess return vs SPY over 10 months of walk-forward paper trading (2025-09-22 → 2026-07-31; 2,008 model ratings → 181 positions; Sharpe 1.82, max drawdown –11.3%), real-cash mirror live since 2026-07. Comparative variants on LightGBM / TabPFN / XGBoost and agentic RAG. Full CI/CD on GitHub Actions; 602-test suite. Methodology and verified figures: eriktaylor.github.io/evidence/portfolio-manager
- **dev-plan (MIT, open-source).** Claude Code skill generating structured development plans with dependency-ordered implementation waves and a weighted scoring model for build sequencing. Design writeup published in Digital Mind.
- **Digital Mind – Technical Series (Medium).** Author of a technical series on frontier AI architectures – mathematical teardowns and worked implementations covering vision transformers, diffusion models, latent attention, and LLM reasoning and memorization.

TECHNICAL SKILLS

Agentic AI & LLM Applications: LLM application development (RAG, agents, tool use, function calling) · Claude / Claude Code (daily use) · Claude Code skill and plugin authoring · Anthropic API (via OpenRouter) · LangChain · LangGraph · LlamaIndex · autonomous agent orchestration · Model Context Protocol (MCP) · multi-modal AI · prompt engineering

Model Evaluation & AI Safety: Frontier-model reasoning failure analysis · evaluation harness design · model risk and validation · benchmarking (accuracy / latency / safety / cost trade-offs) · red-teaming · alignment-informed test design

Frameworks & Languages: Python (expert) · PyTorch · TensorFlow/Keras · Hugging Face Transformers · scikit-learn · LightGBM/XGBoost · FAISS · JavaScript · SQL · OpenCV

MLOps & Infrastructure: AWS · GCP · Docker · CI/CD (GitHub Actions) · Cloudflare (Pages, Workers) · distributed training (DDP, FSDP) · experiment tracking (W&B) · Linux/Ubuntu

Computer Vision: Detection, segmentation, depth estimation · open-vocabulary detection (OWL-ViT, CLIP) · self-supervised vision (DINOv2) · on-device / cascaded inference · quantization (INT8/INT4) · knowledge distillation · ONNX

Domain Knowledge: Medical imaging (MRI, PET, CT) · physics-informed ML · clinical data governance · translational research workflows · edge hardware constraints (memory, latency, power)

STRATEGIC CAPABILITIES

AI Product Strategy & Risk Management: Directing model evaluation and vendor selection · foundation model benchmarking · agentic and algorithmic stress-testing · defining model validation constraints · setting deployment and evaluation criteria across cloud and on-device targets.

Cross-Functional Program Leadership: Technical program management and full product lifecycle ownership · stakeholder management across clinical, legal, regulatory, and engineering teams · external partner and vendor coordination · matrixed team leadership.

Clinical AI & Regulatory Awareness: FDA SaMD landscape (PCCP, GMLP, IEC 62304 – actively developing) · EU AI Act high-risk classification · NIST AI RMF · ISO 42001 · HIPAA · IRB-governed research protocols.

EDUCATION & RESEARCH TRAINING

PhD, Biomedical Engineering – Brown University

Doctoral research in theranostic technologies, integrating advanced medical imaging diagnostics with targeted therapeutics.

NIH Research Fellow, Cardiovascular & Brain MRI – Boston University

Federally funded fellowship developing and evaluating deep learning frameworks for translational medical imaging.

BS, Biomedical Engineering – University of Texas at Austin

Graduate Coursework, Data Science – Harvard University

SELECTED PUBLICATIONS

Selected from 50+ peer-reviewed publications. Full list available on request.

Divani AA, Salazar P, Ikram HA, **Taylor E**, Wilson CM, et al. Non-invasive vagus nerve stimulation improves brain lesion volume and neurobehavioral outcomes in a rat model of traumatic brain injury. *Journal of Neurotrauma*. 2023; 40(13–14): 1481–1494. ★ **Journal Cover Feature**

Taylor EN, Huang N, Lin S, Mortazavi F, Wedeen VJ, et al. Lipid and smooth muscle architectural pathology in the rabbit atherosclerotic vessel wall using Q-space cardiovascular magnetic resonance. *Journal of Cardiovascular Magnetic Resonance*. 2022; 24(1): 74.

Taylor EN, Ding Y, Zhu S, Cheah E, et al. Association between tumor architecture derived from generalized Q-space MRI and survival in glioblastoma. *Oncotarget*. 2017; 8(26): 16296.